Question 1: Siegel's Paradox

Let X_t be the USD/CAD exchange rate, quoted as the price of one US dollar in Canadian dollars, so CAD is the domestic currency and USD is the foreign currency. Today $X_0 = 1.40$ CAD per USD. Over one period the rate either rises by 5% (u = 1.05) or falls by 5% (d = 0.95). Both currencies pay interest of 3% per period: r_CAD = r_USD = 3%.

A Canadian investor can spend X_0 CAD to buy US dollars and deposit it to earn r_USD. At time t=1 the US dollar accounts holds (1 + r_USD), which is worth X_1 (1 + r_USD) in Canadian dollars at that time. Under Q_CAD, the price of any traded asset scaled by the CAD discount factor is a martingale. So today's CAD price equals the expected CAD payoff, discounted at r_CAD:

$$X_0 = \frac{E^{Q_{CAD}}[\mathrm{X}_1(1 + r_{USD})]}{1 + r_{CAD}} \quad \Longrightarrow \quad E^{Q_{CAD}}[\mathrm{X}_1] = X_0 \, \frac{1 + r_{CAD}}{1 + r_{USD}}$$

a. The expected rate X_1 also has to equal the probability-weighted sum of the two outcomes of up and down. Use this to show that p = [(1 + r_CAD)/(1 + r_USD) − d]/(u − d) and write down the expression for (1-p).

b. The CAD forward price of one US dollar is F = X_0(1 + r_CAD)/(1 + r_USD). Compute F, and show that E^Q_CAD[X_t] = F.

c. An American investor quotes the same rate the other way round, as 1/ X_1 US dollars per Canadian dollar, so their forward rate is 1/F. Compute E^Q_CAD[1/X_1] and compare it with 1/F. Explain the difference.

d. Now take the American investor's point of view. Their traded asset is a CAD deposit worth (1/ X_1)(1 + r_CAD)^t in US dollars, and their numeraire is the USD money-market account. Derive the USD risk-neutral probability p' of an up move in X, compute it, and verify that p' = p·u·(1 + r_USD)/(1 + r_CAD).

e. Compute E^Q_USD[1/ X_1] and explain the significance.

f. Compute the Radon–Nikodym derivative defined as dQ_USD/dQ_CAD = p'/p in the up state and (1 − p')/(1 − p) in the down state. Show that in both states it equals X_1/F, the ratio of the two discount factors. Check that its expectation under Q_CAD is 1.

Question 2: Martingales on the Tree

Work in a two-period binomial model with $S_0 = \$100$, $u = 1.20$, $d = 0.90$ and a risk-free rate of $r = 5\%$ per period, so the stock can reach \$120 or \$90 at $t = 1$ etc. Let Q be the risk-neutral measure, under which the stock goes up with probability p each period, and let $\pi = 0.6$ be the "real-world" probability of an up move. Let $M_t = S_t/(1.05)^t$ be the discounted stock price.

1. Determine p and compute $E^Q[S_2 \mid S_1 = \$120]$ and $E^Q[S_2 \mid S_1 = \$90]$. Use them to verify that M is a Q-martingale from $t = 1$ to $t = 2$ at both nodes, and from $t = 0$ to $t = 1$.
2. Use the tower property to compute $E^Q[S_2]$ two ways: directly from the terminal distribution, and by first conditioning on $t = 1$. Is it the same or different and why?
3. Repeat the check in part 1 using $\pi = 0.6$ instead of p. Is M a martingale, a submartingale or a supermartingale under π? What real-world fact about the stock does this reflect?

Question 3: Stopping Times and Optional Stopping

An investor who holds no shares will buy one share at a random time $\tau \in \{0, 1, 2\}$. Let $M_t = S_t/(1.05)^t$ be the discounted stock price, so $M_\tau = S_\tau/(1.05)^\tau$ is the trade price discounted back to $t = 0$, and $M_0 = \$100$. Using the model from Q. 2, consider three trading rules:

- **τ_A:** buy the first time the stock reaches \$120 or more; otherwise buy at $t = 2$.
- **τ_B:** buy the first time the stock falls to \$90 or below; otherwise buy at $t = 2$.
- **τ_C:** buy at the time when the stock is at its *lowest* over $t = 0, 1, 2$ (if tied, the earliest such time).

1. For each of τ_A, τ_B and τ_C, say whether it is a stopping time and justify your answer.
2. For each rule, fill in a table with one row per path (uu, ud, du, dd) showing its Q-probability, τ, the discounted trade price M_τ and their product. Compute $E^Q[M_\tau]$ for each rule.
3. State the optional stopping theorem and use it to explain your results for τ_A and τ_B. What does it imply about trying to time the purchase of a stock based on past prices?
4. Explain why τ_C breaks the theorem, and what the gap between $E^Q[M_\tau]$ and M_0 represents for the buyer.

Assignment 2 — Jaime Corro Galdo'

① X_1: USD/CAD exchange rate (price of one US dollar in canadian dollars)

$X_0 = 1,4$ CAD per USD

$U = 1,05$ $\quad r_{CAD} = r_{USD} = 3\%$

$d = 0,95$

$t = 0 \rightarrow X_0$ CAD to buy 1 US dollar

$t = 1 \rightarrow 1 \cdot (1 + r_{USD})$ USD $= X_1 (1 + r_{USD})$ CAD

Under Q_{CAD} : $X_0 = \dfrac{E^{Q_{CAD}}[X_1(1+r_{USD})]}{1+r_{CAD}} \Rightarrow E^{Q_{CAD}}[X_1] = X_0 \dfrac{1+r_{CAD}}{1+r_{USD}}$

a) $E^{Q_{CAD}}[X_1] = pU X_0 + (1-p) X_0$, where p is the risk-neutral probability of the up state under Q_{CAD}

We know that $X_0 = \dfrac{E^{Q_{CAD}}[X_1(1+r_{USD})]}{1+r_{CAD}} \Leftrightarrow$

~~$\left(\Leftrightarrow X_0 = \dfrac{U X_0 + d X_0}{1+r_{CAD}} \Leftrightarrow 1 + r_{CAD} = U + d \right)$~~

$\Leftrightarrow X_0 [pU + (1-p)d] = X_0 \dfrac{1+r_{CAD}}{1+r_{USD}} \Leftrightarrow p(U-d) + d = \dfrac{1+r_{CAD}}{1+r_{USD}}$

$\Leftrightarrow \left[p = \dfrac{\frac{1+r_{CAD}}{1+r_{USD}} - d}{U-d} = \dfrac{1-0,95}{1,05-0,95} = \dfrac{0,05}{0,1} = \dfrac{1}{20} : \dfrac{1}{10} = \dfrac{10}{20} = \dfrac{1}{2} \right]$

$\left[1 - p = 1 - \left(\dfrac{\frac{1+r_{CAD}}{1+r_{USD}} - d}{U-d} \right) = \dfrac{U - \frac{1+r_{CAD}}{1+r_{USD}}}{U-d} = \dfrac{1,05 - 1}{0,1} = \dfrac{1}{2} \right]$

Q1) CAD forward price of one US dollar $\Rightarrow F = X_0 \frac{(1+r_{CAD})}{1+r_{USD}}$

A forward exchange rate is the rate agreed upon today for exchanging currencies at a specified future date.

There are two strategies:

A] Enter into a forward contract today, fixing the exchange rate at F

B] Purchase USD today, invest it and earn r_{USD} to get to $1 USD in a year. Therefore you need to purchase $\frac{\$1}{1+r_{USD}}$ USD today to end up with $1 USD at year-end.
To do so, you will need to borrow $\frac{\$X_0}{1+r_{USD}}$ CAD today and in one year you will pay back the loan, with Canadian interest.
The amount owed back is $\$X_0 \cdot \frac{1+r_{CAD}}{1+r_{USD}}$ CAD

Because of no arbitrage, the cost of these strategies must be consistent $\Rightarrow \boxed{F = X_0 \frac{(1+r_{CAD})}{(1+r_{USD})} = 1,4 \cdot \frac{1,03}{1,03} = 1,4}$

Now from part a)

$$E^{Q_{CAD}}[X_1] = pX_0u + (1-p)X_0d = X_0[pu + (1-p)d]$$

From part a), we have established:

$$pu + (1-p)d = \frac{1+r_{CAD}}{1+r_{USD}} \Rightarrow \boxed{E^{Q_{CAD}}[X_1] = X_0 \frac{1+r_{CAD}}{1+r_{USD}} = F}$$

c) The American investor wants to know how many USD one CAD costs. Therefore, they use: $Y_t = \frac{1}{X_t}$

p Canadian risk neutral probabilities

$$E^{Q_{CAD}}\left[\frac{1}{X_1}\right] = p\frac{1}{X_1^u} + (1-p)\frac{1}{X_1^d} = \overset{0,5}{p}\frac{1}{1,4(1,05)} + (1-p)\frac{1}{1,4(0,95)} = 0,716$$

From part b), $F = 1,4 \Rightarrow \frac{1}{F} = \frac{1}{1,4} = 0,7143$

The two quantities are not exactly the same, this is because Jensen's inequality, using that $\frac{1}{x}$ is a strictly convex function for positive x, we have:

$$E\left[\frac{1}{X}\right] > \frac{1}{E[X]}$$

The American investor's reciprocal forward rate is still $\frac{1}{F}$. The issue is that we are calculating the expected reciprocal using Q_{CAD}, the probability measure associated with the Canadian numeraire

d) If an American investor wants to buy 1 CAD today and invest it in a Canadian bank, because $X_0 = 1,4$ CAD per USD

$\Rightarrow$ cost of 1 CAD in USD = $\frac{1}{X_0}$ = price today (V_0), after one period: 1 CAD $\times (1+r_{CAD})$

$= \frac{1}{X_1}$ USD $\times (1+r_{CAD})$ = future payoff (V_1)

Now we are going to use that $V_0 = \frac{E^Q[V_1]}{1+r}$

$$\Rightarrow \frac{1}{X_0} = \frac{1}{1+r_{USD}} E^{Q_{USD}}\left[\frac{1+r_{CAD}}{X_1}\right] \quad (1)$$

Now let's define $p' = Q_{USD}$(up move in X). The exchange rate still has the same two outcomes: $X_1^u = uX_0$ and $X_1^d = dX_0$. What changes is the probability assigned to those outcomes

$$E^{Q_{USD}}\left[\frac{1+r_{CAD}}{X_1}\right] = (1+r_{CAD})\left[\frac{p'}{X_0 u} + \frac{(1-p')}{X_0 d}\right]$$

We substitute in (1):

$$\frac{1}{X_0} = \frac{1}{1+r_{USD}} 1+r_{CAD}\left[\frac{p'}{X_0 u} + \frac{(1-p')}{X_0 d}\right]$$

$$\Leftrightarrow \delta = \frac{1+r_{CAD}}{1+r_{USD}}\left[\frac{p'}{u} + \frac{1-p'}{d}\right] \Leftrightarrow \frac{1+r_{USD}}{1+r_{CAD}} = \frac{p'd + u - p'u}{ud} \Leftrightarrow$$

$$\Leftrightarrow \frac{\frac{1+r_{USD}}{1+r_{CAD}} \cdot ud - u}{d-u} = p' \Leftrightarrow p' = \frac{u\left(d\cdot\frac{1+r_{USD}}{1+r_{CAD}} - 1\right)}{d-u}$$

We check:

$$\left[p u \frac{1+r_{USD}}{1+r_{CAD}} = \frac{u\left(\frac{1+r_{CAD}}{1+r_{USD}} - d\right)}{u-d}\frac{1+r_{USD}}{1+r_{CAD}} = \frac{u\left(d - \frac{1+r_{CAD}}{1+r_{USD}}\right)}{d-u}\frac{1+r_{USD}}{1+r_{CAD}} =\right.$$

$$\left. = \frac{u\left(d\frac{1+r_{USD}}{1+r_{CAD}} - 1\right)}{d-u} = p'\right].$$

$$\left[p' = \frac{1{,}05(0{,}95-1)}{0{,}95-1{,}05} = 0{,}525\right]$$

e) We want to calculate $E^{Q_{USD}}\left[\frac{1}{X_1}\right]$. The difference between this part and part c) is the change in probability measure

$$\left.\begin{aligned} X_1^u &= 1{,}4(1{,}05) = 1{,}47 \\ X_1^d &= 1{,}4(0{,}95) = 1{,}33 \end{aligned}\right\} \Rightarrow \begin{aligned} \frac{1}{X_1^u} &= \frac{1}{1{,}47} \\ \frac{1}{X_1^d} &= \frac{1}{1{,}33} \end{aligned}$$

$$E^{Q_{USD}}\left[\frac{1}{X_1}\right] = \frac{p'}{X_1^u} + \frac{(1-p')}{X_1^d} = \frac{0{,}525}{1{,}47} + \frac{0{,}475}{1{,}33} = 0{,}7143$$

Now we obtain the same value as the reciprocal forward rate $\frac{1}{F}$.

We can also see this taking the equation pricing a Canadian-dollar bank deposit in US dollars and $V_0 = \frac{E^Q[V_1]}{1+r}$:

$$\frac{1}{X_0} = \frac{1+r_{CAD}}{1+r_{USD}} E^{Q_{USD}}\left[\frac{1}{X_1}\right] \Leftrightarrow \left[E^{Q_{USD}}\left[\frac{1}{X_1}\right] = \frac{1}{X_0}\frac{1+r_{USD}}{1+r_{CAD}} = \frac{1}{F}\right]$$

8) The Radon-Nikodym derivative tells us how to transform probabilities under one measure into probabilities under another.

$$Z = \frac{dQ_{USD}}{dQ_{CAD}}$$

For any possible state:

$$Z(\text{state}) = \frac{\text{Probability under } Q_{USD}}{\text{Probability under } Q_{CAD}} \quad \text{(ratio of probabilities)}$$

$$Z_U = \frac{p'}{p} = \frac{0,525}{0,5} = 1,05 \quad \text{and} \quad Z_D = \frac{1-p'}{1-p} = \frac{0,475}{0,5} = 0,95$$

Now we verify, $Z \overset{?}{=} \frac{X_1}{F}$

$$\frac{X_1^U}{F} = \frac{1,4(1,05)}{1,4} = 1,05 = Z_U \quad \text{and} \quad \frac{X_1^D}{F} = \frac{1,4 \cdot 0,95}{1,4} = 0,95 = Z_D$$

The ratio $\frac{X_1}{F}$ measures the realised future exchange rate relative to the forward exchange rate.

- If $X_1 > F \Rightarrow$ the USD measure gives the state more weight
- If $X_1 < F \Rightarrow$ " " " " " " less "

Finally, $E^{Q_{CAD}}[Z] = p \underbrace{Z_U}_{\frac{p'}{p}} + (1-p) \underbrace{Z_D}_{\frac{1-p'}{1-p}} = 0,5 \cdot 1,05 + 0,5 \cdot 0,95 = 1$

This makes sense because Z transforms the old probabilities into new probabilities, which must sum to 1.

We can also see that $E^{Q_{USD}}[Y] = E^{Q_{CAD}}[ZY]$, taking $Y = \frac{1}{X_1}$

$$\left[E^{Q_{USD}}\left[\frac{1}{X_1}\right] = E^{Q_{CAD}}\left[Z\frac{1}{X_1}\right] = E^{Q_{CAD}}\left[\frac{1}{F}\right] \right]$$

$$\downarrow Z = \frac{X_1}{F}$$

(2) Two-period binomial model

$S_0 = \$100$, $u = 1,2$, $d = 0,9$ risk-free rate $\Rightarrow r = 5\%$

a) Under the risk-neutral measure, the stock's conditional expected return equals the risk-free rate:

$$E^Q[S_{t+1} | F_t] = (1+r) S_t$$

$$\Rightarrow E^Q[S_1] = p S_0 u + (1-p) S_0 d = (1+r) S_0 \Leftrightarrow pu + (1-p)d = 1+r$$

$$\Leftrightarrow \left[p = \frac{1+r-d}{u-d} \right] \qquad p = \frac{1 + 0,05 - 0,9}{1,2 - 0,9} = \frac{1}{2}$$

Now we want to compute the expected price at time 2 under Q, knowing that the stock has already reached \$120 at time 1.

$$E^Q[S_2 | S_1 = 120] = p\,120 \cdot 1,2 + (1-p)\,120 \cdot 0,9 = 126$$

The same for:

$$E^Q[S_2 | S_1 = 90] = p \cdot 90 \cdot 1,2 + (1-p)\,90 \cdot 0,9 = 94,5$$

Now we have the process $M_t = \frac{S_t}{1,05^t}$, is $E^Q[M_2 | S_1] = M_1$?

For the upper node: $\left[E^Q[M_2 | S_1 = 120] = \frac{E^Q[S_2 | S_1 = 120]}{(1,05)^2} = \frac{126}{1,05^2} = \frac{120}{1,05} = M_1 \right]$

For the lower node $\left[E^Q[M_2 | S_1 = 90] = \frac{E^Q[S_2 | S_1 = 90]}{(1,05)^2} = \frac{94,50}{1,05^2} = \frac{90}{1,05} = M_1 \right]$

Now between $t=0$ and $t=1$, $\left[E^Q[M_1 | S_0 = 100] = \frac{E^Q[S_1 | S_0 = 100]}{1,05} = \frac{0,5 \cdot 100 \cdot 1,2 + 0,5 \cdot 100 \cdot 0,9}{1,05} = \right.$

$$\left. = \frac{105}{1,05} = 100 = M_0 \right]$$

So we have verified that M is a Q-martingale from $t=1$ to $t=2$ and from $t=0$ to $t=1$

b) The Tower property states that averaging in stages = averaging all at once:

$$E_0[E_1[X]] = E_0[X]$$

Let's compute $E^Q[S_2]$

i) Directly from the terminal distribution

$$E^Q[S_2] = p^2 S_0 u^2 + 2p(1-p) S_0 ud + (1-p)^2 S_0 d^2 = 110{,}25$$

↓ Binomial sum

ii) By first conditioning on $t=1$

$$E^Q[E^Q[S_2 | F_1]] = p\,126 + (1-p)\,94{,}5 = 110{,}25$$

$$E^Q[S_2 | S_1 = 120] = p(120u) + (1-p)(120d) = 126$$

$$E^Q[S_2 | S_1 = 90] = p\,90u + (1-p)\,90d = 94{,}5$$

Both give the same result, which makes sense, because in method i) we consider the probability of every possible path from today until $t=2$.

In method ii) we group those paths according to their first movement.

For the upper node, we consider uu and ud

For the lower node, we consider du and dd.

Then we average those two groups.

Another interesting relationship from a) is that $E^Q[S_0] = S_0(1+r)^2$

c) $E^\pi[S_2 | S_1 = 120] = 0{,}6 \cdot 120 \cdot u + 0{,}4 \cdot 90 \cdot d = 129{,}6$

$E^\pi[S_2 | S_1 = 90] = 0{,}6 \cdot 90 \cdot u + 0{,}4 \cdot 90 \cdot d = 97{,}2$

Now we have the process $M_t = \frac{S_t}{1{,}05^t}$, is $E^\pi[M_2 | S_1] = M_1$?

$$E^{\pi}[M_2 \mid S_1 = 120] = \frac{E^{\pi}[S_2 \mid S_1 = 120]}{(1{,}05)^2} = \frac{124{,}6}{1{,}05^2} = \frac{123{,}042857}{1{,}05} > M_1 = \frac{120}{1{,}05}$$

$$E^{\pi}[M_2 \mid S_1 = 90] = \frac{E^{\pi}[S_2 \mid S_1 = 90]}{(1{,}05)^2} = \frac{97{,}2}{1{,}05^2} = \frac{92{,}571428}{1{,}05} > M_1 = \frac{90}{1{,}05}$$

$$E^{\pi}[M_1 \mid S_0 = 100] = \frac{E^{\pi}[S_1 \mid S_0 = 100]}{1{,}05} = \frac{0{,}6 \cdot 100 + 0{,}4 \cdot 100 \cdot d}{1{,}05} = \frac{108}{1{,}05} = 102{,}857142 \quad V \quad M_0 = 100$$

We conclude that $E_t[M_{t+1}] \geq M_t \Rightarrow M_t$ is a submartingale

This is because under real-word probabilities, investors may expect a return different from the risk-free rate.

$$E^{\pi}[S_2] = 0{,}6 \cdot 120 + 0{,}4 \cdot 90 = 108$$

$$\text{Expected return} = \frac{E^{\pi}[S_1]}{100} - 1 = 0{,}08 > r = 0{,}05$$

(3) Discounted stock price: $M_t = \frac{S_t}{(1{,}05)^t}$

Trade price discounted back to $t=0$: $M_\tau = \frac{S_\tau}{1{,}05^\tau}$

$M_0 = \$100$

τ_A: Buy the first time the stock reaches \$120 or more, otherwise at $t=2$

τ_B: " " " " " " falls to \$90 or below; " " "

τ_C: buy at the time when the stock is at its lowest over $t=0,1,2$

a) The definition of a stopping time is a random time $\tau \in \{0, 1, \ldots, n\}$ such that for every t, whether $\tau = t$ can be decided from the path up to t.

Based on the definition, we can see that τ_A and τ_B are stopping times, because at each time you can see whether S_t

has cross the threshold.

On the other hand, τ_C is not a stopping time because at $t=0$ you don't what will happen, you can't decide what you should stop

a) First we are going to determine the possible path for each path.

For τ_A the stock reaches 120 at $t=1 \Rightarrow$ paths uu and ud

For τ_B the stock falls to 90 at $t=1 \Rightarrow$ paths du and dd

For τ_C, for paths uu, ud $\Rightarrow t=0$

for path dd $\Rightarrow t=2$

for path du $\Rightarrow t=1$

Now we calculate the discounted purchase price

$M_\tau = \frac{S_\tau}{1{,}05^\tau}$ and $Q \times M_\tau$ and we fill the tables

τ_A

Path	Q-prob	τ	M_τ	$Q \times M_\tau$
uu	0,25	1	114,286	28,571
ud	0,25	1	114,286	28,571
du	0,25	2	97,959	24,4898
dd	0,25	2	73,4694	18,3673

τ_B

Path	Q-prob	τ	M_τ	$Q \times M_\tau$
uu	0,25	2	130,6122	32,653
ud	0,25	2	97,9592	24,4898
du	0,25	1	85,7143	21,4286
dd	0,25	1	85,7142	21,4286

τ_C

Path	Q-prob	τ	M_τ	$Q \times M_\tau$
uu	0,25	0	100	25
ud	0,25	0	100	25
du	0,25	0	85,7143	21,4286
dd	0,25	2	73,4694	18,3673

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The difference is that for strategies A and B

$E^Q[M_{\tau_A}] = 100 = E^Q[M_{\tau_B}]$

but for strategy C, $E^Q[M_{\tau_C}] = 89{,}7959$

So the difference between C and buying immediately is

$100 - 89{,}7959 = 10{,}2041$

It appears cheaper, but requires knowledge of future prices

c) The optional stopping theorem states that if M is a martingale and τ is a stopping time with $\tau \leq n$, then

$$E[M_\tau] = M_0$$

For τ_A and τ_B we obtained $E^Q[M_{\tau_A}] = 100 = E[M_{\tau_B}]$

and $M_0 = 100$, so the results agree with the theorem.

This implies that choosing a purchase time using only available information cannot lower the expected discounted purchase cost of a Q-martingale.

d) τ_C fails one of the theorem's assumptions: the stopping-time condition, because knowing that today's price is the lowest requires knowing what happens tomorrow.

Strategy C has a lower expected discounted purchase price than buying immediately, for a buyer, a lower purchase is desirable. That difference $(M_0 - E^Q[M_{\tau_C}])$ represents the saving that would be possible if the investor knew in advance which date would have the lower price, but is not a realistic profit.